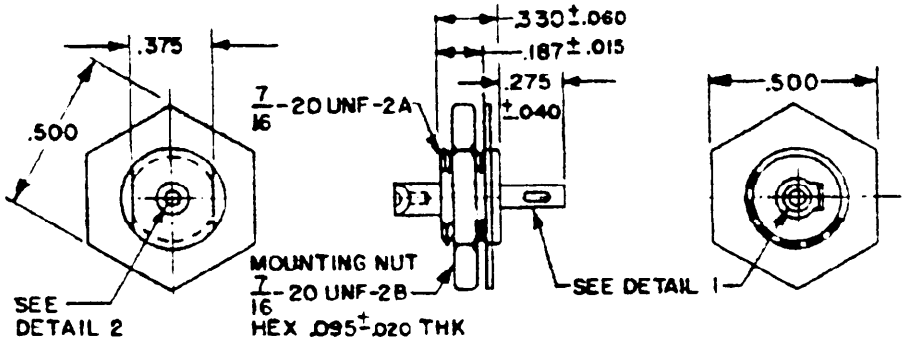
MIL-C-10950/4D
8 July 1965
SUPERSEDING
MIL-C-10950/4C
7 October 1963

MILITARY SPECIFICATION SHEET

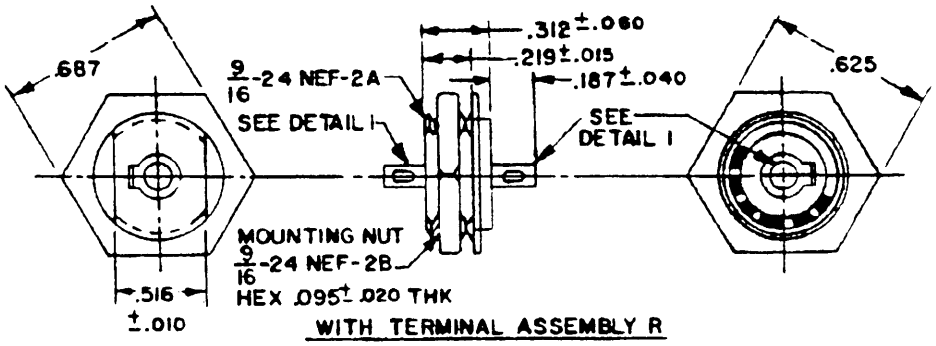
CAPACITORS, FIXED, MICA DIELECTRIC, BUTTON STYLE (FEEDTHRU) STYLE CB50

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-10950.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

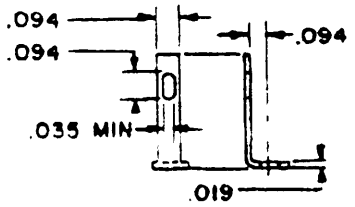


WITH TERMINAL ASSEMBLY L

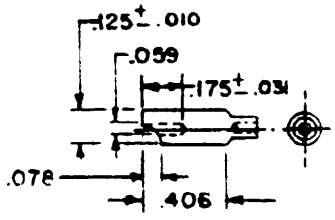


WITH TERMINAL ASSEMBLY R

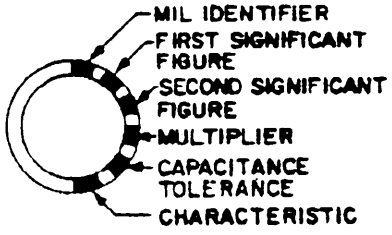
INCHES	MM
.005	.13
.010	.25
.015	.38
.019	.48
.020	.51
.031	.79
.035	.89
.040	1.02
.059	1.50
.060	1.52
.078	1.98
.094	2.39
.095	2.41
.125	3.18
.175	4.45
.187	4.75
.219	5.56
.275	6.99
.312	7.92
.330	8.38
.375	9.53
.406	10.31
.500	12.70
.516	13.11
.625	15.88
.687	17.45



DETAIL 1



DETAIL 2



COLOR-CODE MARKING

NOTES:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are .xxx±.005 (.13 mm).
4. All dimensions are exclusive of sealing compound and color code.
5. Color-coded section shall not exceed 240°.

FSC 5910

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TABLE I. Style CB50.

Part number 1/	Capacitance	DC rated voltage	Characteristic available at capacitance tolerance				Terminal assemblies
			±1% (F)	±2% (G)	±5% (J)	±10% (K)	
	pf	Volts					
CB50-B050K	5	500				B	L, R
CB50-D100K	10	500				D	L, R
CB50-D120K	12	500				D	L, R
CB50-D150K	15	500				D	L, R
CB50-D180K	18	500				D	L, R
CB50-D220-	22	500			D	D	L, R
CB50-D270-	27	500			D	D	L, R
CB50-D330-	33	500			D	D	L, R
CB50-D390-	39	500			D	D	L, R
CB50-D470-	47	500			D	D	L, R
CB50--560-	56	500	D, E	D, E	D, E	D	L, R
CB50--680-	58	500	D, E	D, E	D, E	D	L, R
CB50--820-	82	500	D, E	D, E	D, E	D	L, R
CB50--101-	100	500	D, E	D, E	D, E	D	L, R
CB50--121-	120	500	D, E	D, E	D, E	D	L, R
CB50--151-	150	500	D, E	D, E	D, E	D	L, R
CB50--181-	180	500	D, E	D, E	D, E	D	L, R
CB50--221-	220	500	D, E	D, E	D, E	D	L, R
CB50--271-	270	500	D, E	D, E	D, E	D	L, R
CB50--331-	330	500	D, E	D, E	D, E	D	L, R
CB50--391-	390	500	D, E	D, E	D, E	D	L, R
CB50--471-	470	500	D, E	D, E	D, E	D	L, R
CB50--561-	560	500	D, E	D, E	D, E	D	L, R
CB50--681-	680	500	D, E	D, E	D, E	D	L, R
CB50--821-	820	500	D, E	D, E	D, E	D	L, R
CB50--102-	1,000	500	D, E	D, E	D, E	D	L, R
CB50--122-	1,200	500	D, E	D, E	D, E	D	L, R
CB50--152-	1,500	500	D, E	D, E	D, E	D	L, R

1/ Complete part number will include additional additional symbols to indicate terminal assembly, characteristic, and capacitance tolerance.

REQUIREMENTS:

Dimensions and configuration:

Case: See figure.

Terminals: See figure.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC rated voltage: 500 volts.

Dissipation factor: See figure 1 of MIL-C-10950.

Characteristics: See table I.

Symbol	Temperature coefficient	Capacitance drift
B	Not specified	Not specified
D	-100 to +100	0.3 percent or 0.3 pf whichever is greater.
E	-20 to +100	± (0.1 percent + 0.1 pf).

Operating temperature range: -55° C to 85° C.

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Dielectric withstanding voltage:
Method 301 of MIL-STD-202.
200 percent of rated dc voltage for not less than 1 second, nor more than 5 seconds.

Barometric pressure:
Method 105 of MIL-STD-202, condition C (70,000 ft.)
Test potential - 125 percent of rated dc voltage for not less than 1 second nor more than 5 seconds. Surge current shall not exceed 5 ma.

Insulation resistance:
Method 302 of MIL-STD-202, condition A or B.
At 25° C - 50,000 megohms minimum.
At 85° C - 15,000 megohms minimum.

Capacitance:
Method 305 of MIL-STD-202.
Measurement frequency
Capacitance values of 1000 pf or less - 1 MHz ±100 kHz.
Capacitance values of greater than 1000 pf - 1 kHz ±100 Hz.

Shock (medium impact):
Method 205 of MIL-STD-202, condition C (50G).

Vibration:
Method 204 of MIL-STD-202, condition B.

Moisture resistance:
Test method 106 of MIL-STD-202.
Insulation resistance:
Not less than 500 megohms.
Capacitance change:
Within 1 percent of the initial value or 0.5 pf, whichever is greater.
DF - 200 percent of initial requirement.

Temperature cycling and immersion:
Temperature cycling:
Test method 102 of MIL-STD-202, condition D.
Immersion:
Test method 104 of MIL-STD-202, condition A.
Insulation resistance:
Not less than 5000 megohms.
Capacitance change:
Within 1 percent of the initial value or 0.5 pf, whichever is greater.

Life:
2000 ±12 hours at 85° $^{+3°}_{-0°}$ C.
Insulation resistance:
Greater than 50,000 megohms.
Capacitance change:
Within 1 percent of the initial value or 0.5 pf, whichever is greater.

Temperature coefficient and capacitance drift:
Within the limits specified for the applicable characteristic.

Torque: In accordance with MIL-C-10950.
Mounting - normal mounting means.
Load - 4 ounce-inches.

NOTES:
1. Conversion of dash number to part numbers is as shown in table II.
2. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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mating of the sets in parallel, power exchange between the sets

TABLE II. Dash number to part number conversion.

Dash number in MIL-C-10950/4C				Converted to part number 1/ in MIL-C-10950/4D
Characteristic D		Characteristic E		
Terminal assembly				
L	R	L	R	
0001	0002	0003	0004	CB50--560F
0005	0006	0007	0008	CB50--680F
0009	0010	0011	0012	CB50--820F
0013	0014	0015	0016	CB50--101F
0017	0018	0019	0020	CB50--121F
0021	0022	0023	0024	CB50--151F
0025	0026	0027	0028	CB50--181F
0029	0030	0031	0032	CB50--221F
0033	0034	0035	0036	CB50--271F
0037	0038	0039	0040	CB50--331F
0041	0042	0043	0044	CB50--391F
0045	0046	0047	0048	CB50--471F
0049	0050	0051	0052	CB50--561F
0053	0054	0055	0056	CB50--681F
0057	0058	0059	0060	CB50--821F
0061	0062	0063	0064	CB50--102F
0065	0066	0067	0068	CB50--122F
0069	0070	0071	0072	CB50--152F
0073	0074	0075	0076	CB50--560G
0077	0078	0079	0080	CB50--680G
0081	0082	0083	0084	CB50--820G
0085	0086	0087	0088	CB50--101G
0089	0090	0091	0092	CB50--121G
0093	0094	0095	0096	CB50--151G
0097	0098	0099	0100	CB50--181G
0101	0102	0103	0104	CB50--221G
0105	0106	0107	0108	CB50--271G
0109	0110	0111	0112	CB50--331G
0113	0114	0115	0116	CB50--391G
0117	0118	0119	0120	CB50--471G
0121	0122	0123	0124	CB50--561G
0125	0126	0127	0128	CB50--681G
0129	0130	0131	0132	CB50--821G
0133	0134	0135	0136	CB50--102G
0137	0138	0139	0140	CB50--122G
0141	0142	0143	0144	CB50--152G
0145	0146			CB50--220J
0147	0148			CB50--270J
0149	0150			CB50--330J
0151	0152			CB50--390J
0153	0154			CB50--470J

1/ See footnote at end of table.

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TABLE II. Dash number to part number conversion (continued).

Dash number in MIL-C-10950/4C				Converted to part number 1/ in MIL-C-10950/4D
Characteristic D		Characteristic E		
Terminal assembly				
L	R	L	R	
0155	0156	0157	0158	CB50--560J
0159	0160	0161	0162	CB50--680J
0163	0164	0165	0166	CB50--820J
0167	0168	0169	0170	CB50--101J
0171	0172	0173	0174	CB50--121J
0175	0176	0177	0178	CB50--151J
0179	0180	0181	0182	CB50--181J
0183	0184	0185	0186	CB50--221J
0187	0188	0189	0190	CB50--271J
0191	0192	0193	0194	CB50--331J
0195	0196	0197	0198	CB50--391J
0199	0200	0201	0202	CB50--471J
0203	0204	0205	0206	CB50--561J
0207	0208	0209	0210	CB50--681J
0211	0212	0213	0214	CB50--821J
0215	0216	0217	0218	CB50--102J
0219	0220	0221	0222	CB50--122J
0223	0224	0225	0226	CB50--152J
Characteristic B		Characteristic D		
0227	0228			CB50--B050K
		0229	0230	CB50--D100K
		0231	0232	CB50--D120K
		0233	0234	CB50--D150K
		0235	0236	CB50--D180K
		0237	0238	CB50--D220K
		0239	0240	CB50--D270K
		0241	0242	CB50--D330K
		0243	0244	CB50--D390K
		0245	0246	CB50--D470K
		0247	0248	CB50--D560K
		0249	0250	CB50--D680K
		0251	0252	CB50--D820K
		0253	0254	CB50--D101K
		0255	0256	CB50--D121K
		0257	0258	CB50--D151K
		0259	0260	CB50--D181K
		0261	0262	CB50--D221K
		0263	0264	CB50--D271K
		0265	0266	CB50--D331K

1/ See footnote at end of table.

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TABLE II. Dash number to part number conversion (continued).

Dash number in MIL-C-10950/4C				Converted to part number 1/ in MIL-C-10950/4D
Characteristic B		Characteristic D		
Terminal assembly				
L	R	L	R	
		0267	0268	CB50--D391K
		0269	0270	CB50--D471K
		0271	0272	CB50--D561K
		0273	0274	CB50--D681K
		0275	0276	CB50--D821K
		0277	0278	CB50--D102K
		0279	0280	CB50--D122K
		0281	0282	CB50--D152K

1/ The complete part number shall include symbols for terminal assembly and characteristic, as applicable.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Navy - AS, CG, OS

Preparing activity:
Navy - EC

(Project 5910-0847-5)

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applicable for the mode II, size 5 set